Craftax runs compared: achievements and what unlocks them

The masked family (B1-B3) and the plain baselines: 1.1M env steps, size50m, the same 30 fixed worlds (tools/death_eval.py). Every family run reads the game's rules and counts as cheating. 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds.

The two plain runs are honest: vanilla DreamerV3 4.63 ± 0.22 and plain DreamerV3 on the fixed code 4.77 ± 0.28. Every other row reads the game's action rules (craftax_valid, a transcription of each action's precondition) and is counted as cheating: B1 gets the 43 validity flags as input (6.30 ± 0.33), B2 masks impossible actions (8.40 ± 0.54; fixed RSSM-2 8.27 ± 0.44, three seeds 8.27 / 7.70 / 9.03, mean 8.33), and B3 learns the mask (5.97 ± 0.28), but still needs the rules in its setup. All family runs also train RSSM-2 on hand-built map labels and use the tech-tree potential shaping, both game knowledge in their own right.

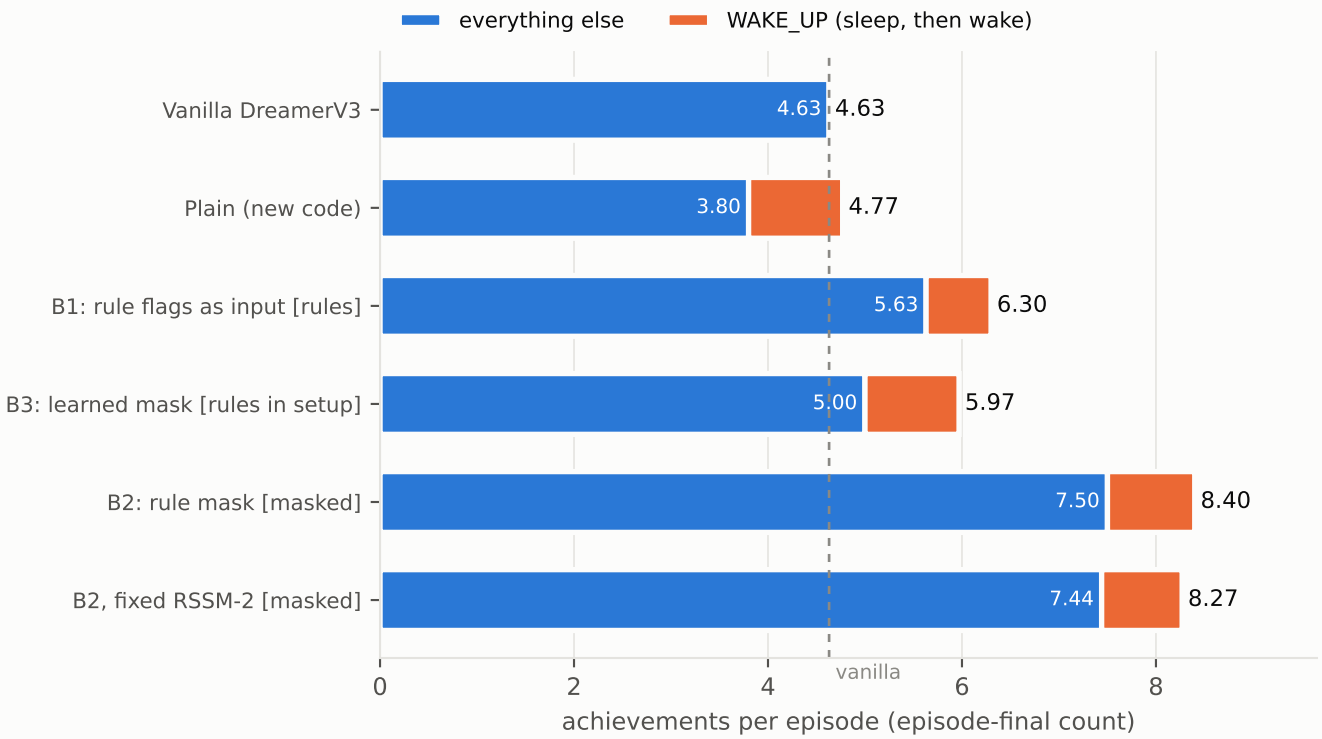
The mask is worth +2.63 ± 0.38 achievements against the same model without it (B2 vs the honest map, fixed code, three seeds), and +2.90 ± 0.26 under the v1.2 manager (60 worlds): it is the largest single gain in the project, and it is bought with the rules.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
B1: rule flags as input [rules]	6.30 ± 0.33	5.63	295	2.39%	93%	53%	7%	27%	77%
B3: learned mask [rules in setup]	5.97 ± 0.28	5.00	269	2.24%	73%	33%	10%	7%	77%
B2: rule mask [masked]	8.40 ± 0.54	7.50	282	3.32%	93%	73%	37%	60%	47%
B2, fixed RSSM-2 [masked]	8.27 ± 0.44	7.44	274	3.32%	97%	80%	40%	57%	67%

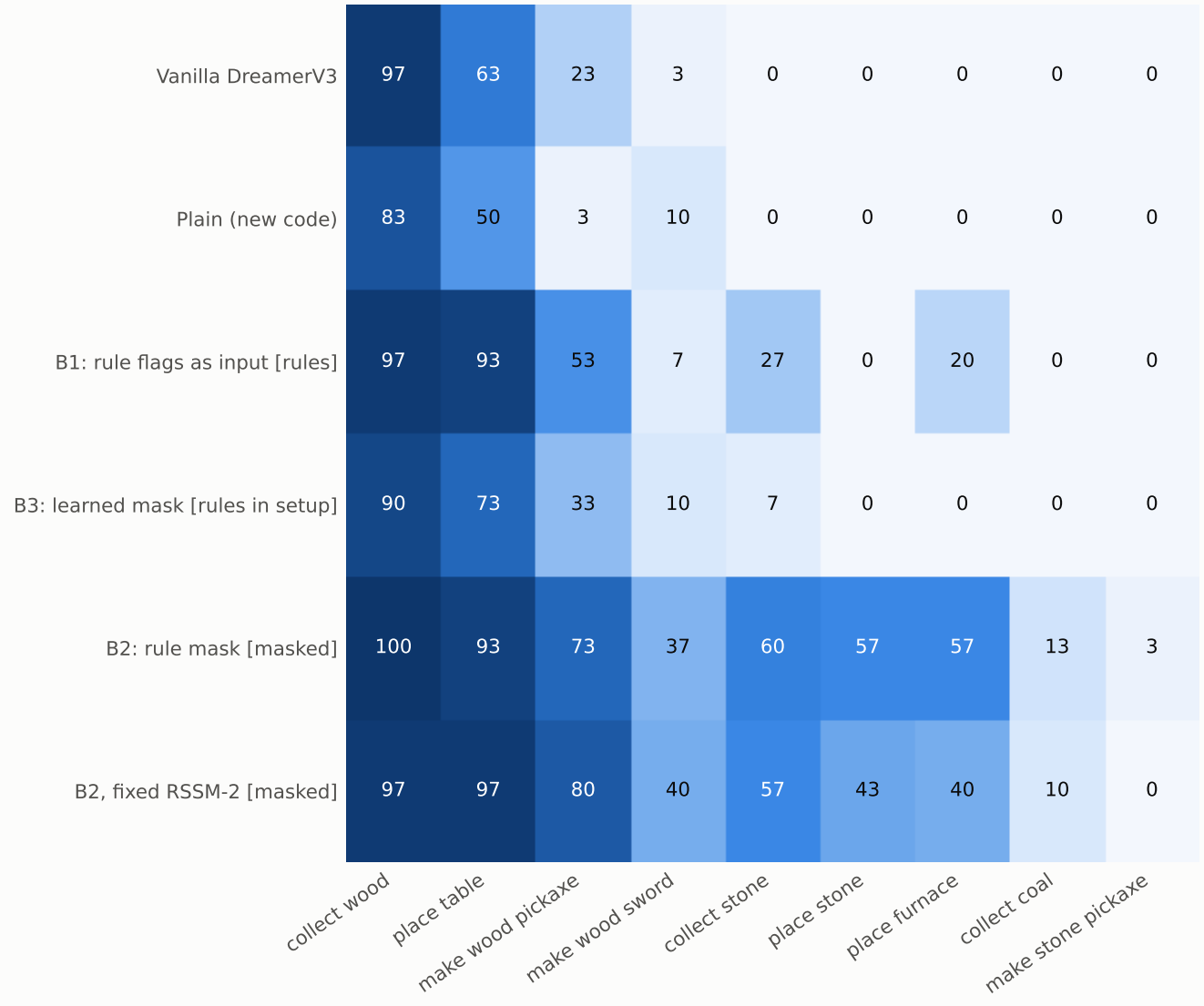
ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed, so seed variation is not included). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. [masked] = impossible actions masked by the rules; [rules] = the rule flags fed to the encoder; [rules in setup] = mask learned from experience, but the env must emit the rule flags to build it, and the basic actions and NOOP are hand-picked. All [..] rows count as cheating. Vanilla: the original checkpoint (code before the 2026-10 fixes), re-scored. Plain: the fixed code, `--configs craftax size50m --env.craftax.survival none`. B1, B2 and B3 ran on the code before the 2026-10-11 RSSM-2 fixes; 'fixed RSSM-2' was retrained after them. B1-B3 sit on the honest-map configuration (RSSM-2 map target, potential shaping); its own runs are on the rssm2 branch.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

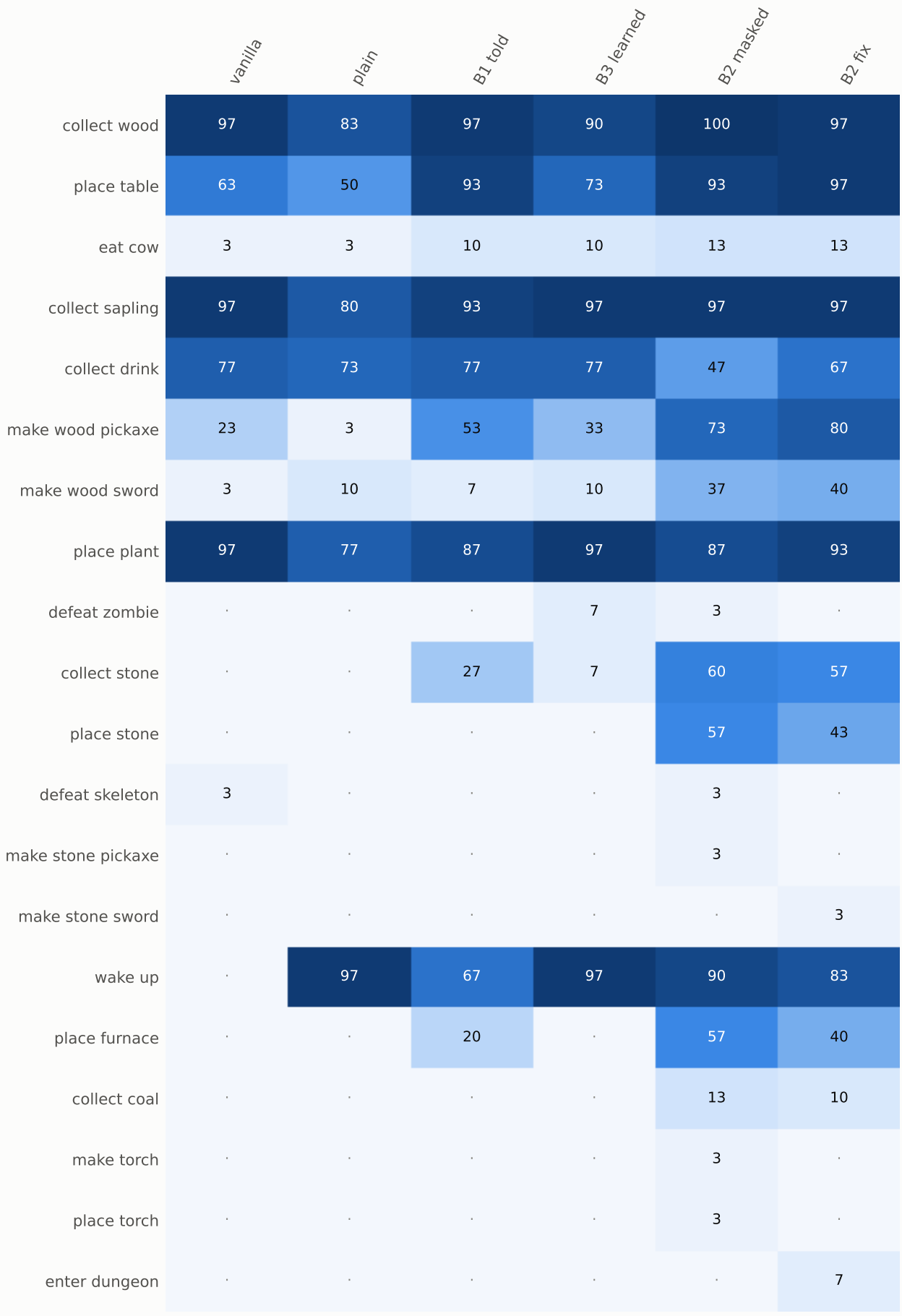


Unlock rate (% of episodes), early tech tree in dependency order



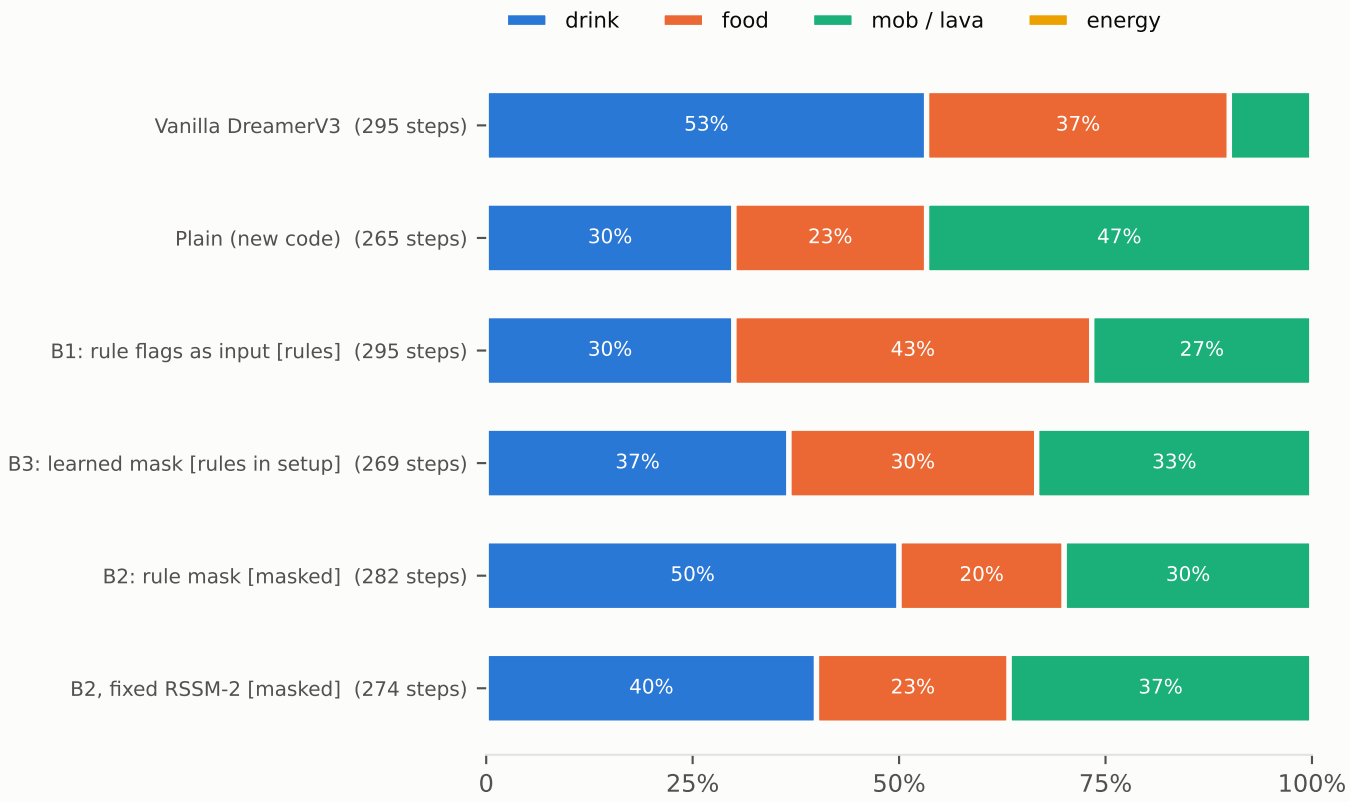
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 47 of 67 were never unlocked by any run.



Cause of death and lifespan

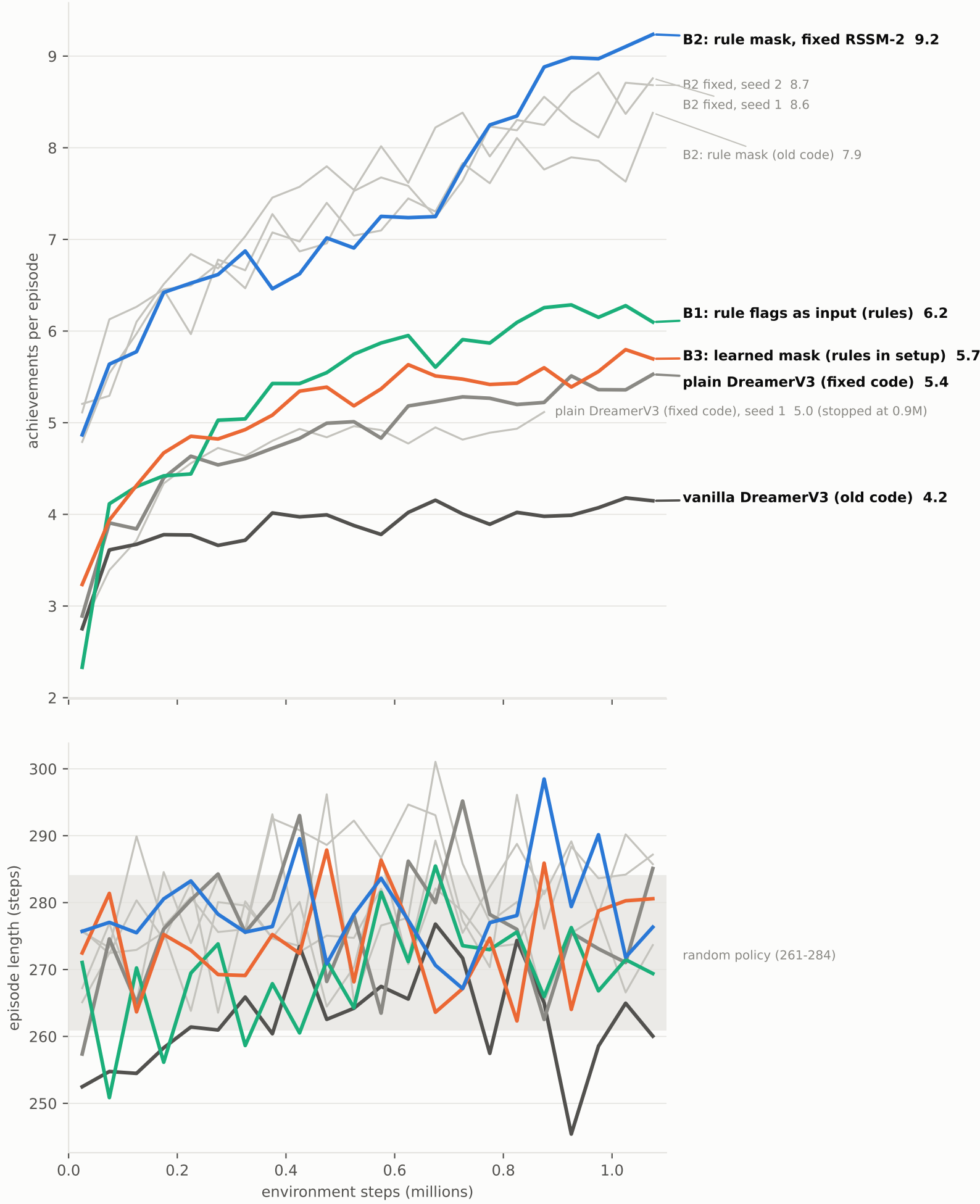
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

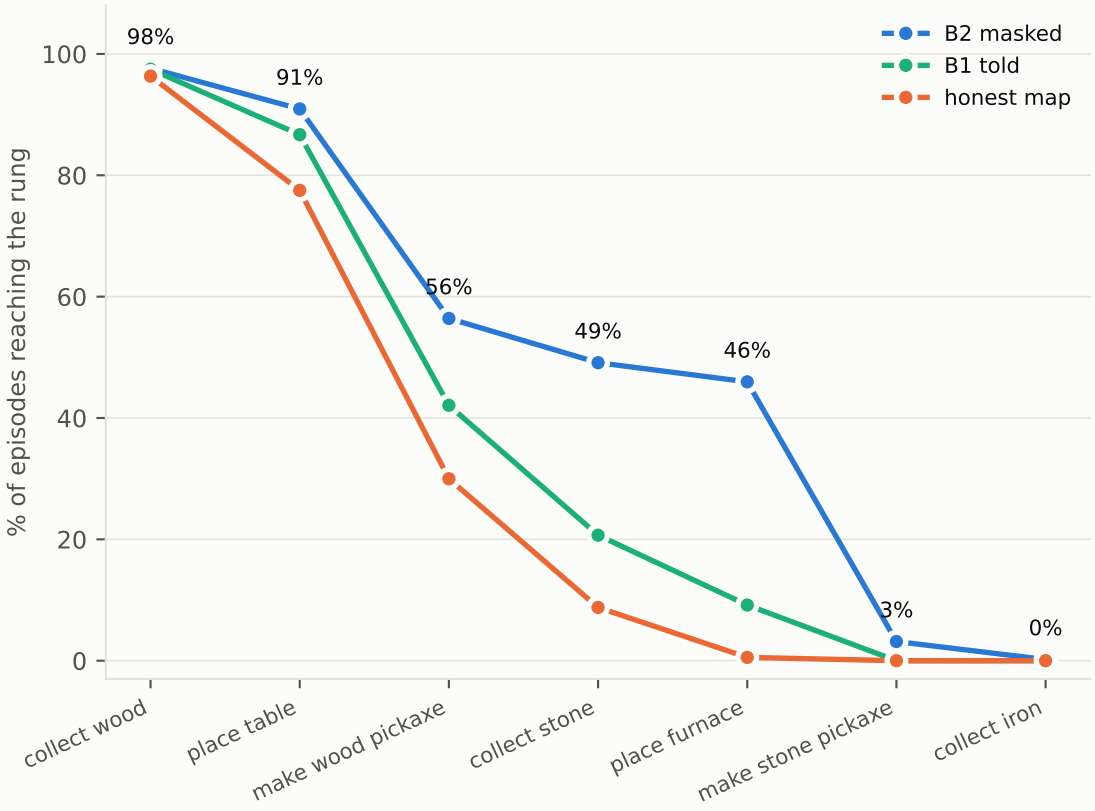
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Craftax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

0. The whole family counts as cheating

craftax_valid.valid_actions transcribes every action's precondition from the game's code. B2 reads it while acting, B1 feeds it to the encoder, and B3, which learns its mask from the world model's counterfactual, still needs the env to emit the flags before the head is built, exempts six hand-picked basic actions and treats action 0 as NOOP. The published 1M-step Craftax baselines use none of this. No new work goes on this branch.

1. Masking impossible actions is a large, robust gain

B2 beats the same model without the mask by +2.63 ± 0.38 (fixed code, three-seed mean; +2.23 on the old code) on the same 30 worlds. Under the v1.2 manager removing the mask costs +2.90 ± 0.26 over 60 worlds. Buttons that were never pressed where they would work (the wood pickaxe, the furnace, placing stone) get pressed once a useless press can no longer teach the agent that the button is bad: valid-action suppression (docs/entropy-and-action-suppression.md).

2. Telling is not masking

B1 sees the same 43 flags as input and scores 6.30, level with the unmasked honest map (6.17) on the fixed worlds; the original evaluation had it 1.0 ahead. Knowing an action is possible does not stop the gradient from pushing it down where it is not.

3. The learned mask failed

B3 (5.97) is level with the honest map and 2.4 below B2. Its labels came from the world model's own latent and its loss trained that latent, so the two fed back on each other: 87% of its 'did something' labels disagreed with the rules. B3-fix (detached, observation-grounded labels) was stopped at 254k steps: its mask let through only 45% of the possible special actions, because the label cannot tell "does nothing" from "not learned yet". Not scored.

4. The RSSM-2 fixes did not move B2

Retrained after the 2026-10-11 fixes, B2 scores 8.27 against 8.40; its three seeds (8.27 / 7.70 / 9.03) differ by 1.3, more than the fix did.

5. What is left

A knowledge-free mask is the only open idea here: block an action only when a world-model ensemble agrees it does nothing, so an effect not yet learned stays pressable. It would go on a new family branch, without valid_obs.